

In this problem set, the questions marked with a $(\star)$ are required for students enrolled in 583 and bonus for students enrolled in 483.

1. Lorentz Transformation

- (a) (2 points) The coordinate transformation for an observer moving at a constant velocity V_x relative to a stationary observer is given by

$$\begin{aligned}t' &= \gamma_x(t - V_x x) \\x' &= \gamma_x(x - V_x t) \\y' &= y \\z' &= z.\end{aligned}\tag{1}$$

How can we write this as a 4×4 Lorentz transformation matrix?

Solution:

$$\Lambda = \begin{pmatrix} \gamma_x & -V_x \gamma_x & 0 & 0 \\ -V_x \gamma_x & \gamma_x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}\tag{2}$$

- (b) (5 points) More generally, suppose we have

$$\begin{aligned}t' &= \mu t + \nu x \\x' &= \sigma t + \gamma x \\y' &= \rho y \\z' &= \lambda z.\end{aligned}\tag{3}$$

How can we write this as a 4×4 Lorentz transformation matrix?

Solution:

$$\Lambda = \begin{pmatrix} \mu & \nu & 0 & 0 \\ \sigma & \gamma & 0 & 0 \\ 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & \lambda \end{pmatrix}\tag{4}$$

2. Coordinate Transformations Let

$$x = r \cos \theta, \quad y = r \sin \theta.$$

- (a) (2 points) Compute $\partial x/\partial r$, $\partial x/\partial \theta$, $\partial y/\partial r$, and $\partial y/\partial \theta$.
- (b) (2 points) Use the chain rule to express $\partial/\partial r$ in terms of $\partial/\partial x$ and $\partial/\partial y$.
- (c) (2 points) Interpret $\partial/\partial r$ geometrically.

Solution:

$$\begin{aligned}\frac{\partial x}{\partial r} &= \cos \theta, & \frac{\partial x}{\partial \theta} &= -r \sin \theta, \\ \frac{\partial y}{\partial r} &= \sin \theta, & \frac{\partial y}{\partial \theta} &= r \cos \theta.\end{aligned}$$

By the chain rule,

$$\frac{\partial}{\partial r} = \cos \theta \frac{\partial}{\partial x} + \sin \theta \frac{\partial}{\partial y}.$$

Geometrically, $\partial/\partial r$ points in the radial direction and measures change along outward-pointing rays from the origin.

3. Differentials and Geometry Given the scalar function $f(x, y) = x^2 y$:

- (a) (3 points) Compute the differential df .
- (b) (2 points) Interpret df geometrically.

Solution: The differential is

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 2xy dx + x^2 dy.$$

Geometrically, df represents the linear (first-order) change in f under an infinitesimal displacement (dx, dy) . It is a covector that maps a small displacement vector to the corresponding change in the scalar field.

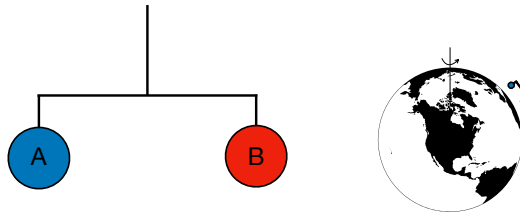
4. (4 points) Change of Basis Write the 2×2 rotation matrix $R(\theta)$ and show how vector components transform.

Solution: The rotation matrix is

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Components transform as $v^i = R^i_j v^j$. The vector itself is unchanged; only its coordinate description depends on the basis.

5. **Eötvös Experiment** Consider a torsion balance consisting of two equal inertial masses mounted at opposite ends of a light horizontal rod and suspended from a thin fiber. The apparatus is fixed to the surface of the rotating Earth at latitude θ . See figure for details.



- (a) (3 points) Define a convenient coordinate system for analyzing the torsion balance. Using this coordinate system, state the condition that must be satisfied for the torsion balance to be in mechanical equilibrium.

Solution: Let the z -axis run along the wire from which the pendulum hangs and the y -axis run along the bar connecting the two test masses. Mechanical equilibrium states that there should be no torque in the x -direction. The torque experienced by one test mass comes from both the gravitational force and the centrifugal force due to the Earth's rotation: $\tau_i = \vec{r}_i \times (\vec{F}_{i,g} + \vec{C}_i)$ where $\vec{C}_i = m_{i,I} \vec{a}_{c,z}$ and $m_{i,I}$ is the inertial mass of test mass i . In the coordinate system defined, the condition for mechanical equilibrium is given by $r_1 m_{1,G} g - C_{1,z} = r_2 m_{2,G} g - C_{2,z}$ or

$$\begin{aligned} \frac{r_1}{r_2} &= \frac{m_{2,G} g - m_{2,I} a_{c,z}}{m_{1,G} g - m_{1,I} a_{c,z}} = \frac{m_{2,G} - m_{2,I} a_{c,z}/g}{m_{1,G} - m_{1,I} a_{c,z}/g} \\ &\simeq \frac{m_{2,G}}{m_{1,G}}, \end{aligned} \tag{5}$$

where in the second line we use that $\vec{a}_c \ll \vec{g}$, i.e. the centrifugal acceleration coming from the rotation of the planet is significantly smaller than the gravitational acceleration from the Earth. Here, $m_{i,G}$ is the gravitational mass of test mass i .

- (b) (3 points) Determine the torque on the torsion balance arising from the Earth's rotation. Your answer should involve the Earth's angular velocity Ω , the latitude θ , and the relevant geometric parameters of the apparatus.

Solution: From the previous part, we saw that mechanical equilibrium required there to be no torque in the x -direction, but we can still have a torque in the z -direction due to the rotation of the Earth. This torque is given by

$$\begin{aligned}\tau_z &= r_2 m_{2,I} a_{c,x} - r_1 m_{1,I} a_{c,x} = r_2 a_{c,x} \left(\frac{m_{2,I}}{m_{2,G}} m_{1,G} - m_{1,I} \right) \\ &= r_2 a_{c,x} m_{1,G} \left(\frac{m_{2,I}}{m_{2,G}} - \frac{m_{1,I}}{m_{1,G}} \right),\end{aligned}\tag{6}$$

where we use the result of part (a) in the first line. The centrifugal force is given by $C = m_I a_c = m_I R_\oplus \Omega^2 \cos \theta$ for an experiment at latitude θ and $R_\oplus$ is the Earth's radius. Therefore, the x -component of the centripetal acceleration is $a_{c,x} = R_\oplus \Omega^2 \sin \theta \cos \theta$ and

$$\tau_z = r_2 m_{1,G} R_\oplus \Omega^2 \sin \theta \cos \theta \left(\frac{m_{2,I}}{m_{2,G}} - \frac{m_{1,I}}{m_{1,G}} \right).\tag{7}$$

- (c) (2 points) What is the optimal latitude to perform this experiment? In other words, at what latitude would one expect to measure the largest potential torque?

Solution: From part (c), we see that the torque is proportional to $\tau \sim \sin \theta \cos \theta = \sin 2\theta$, which is maximized when $\theta = \pi/4$ or 45° . Note that this is close to the location of Seattle (47°), which is where the Eöt-Wash experiment resides.

- (d) (2 points) State the condition required for the weak equivalence principle

to be satisfied for the two test masses in this experiment.

Solution: The weak equivalence principle is satisfied when the inertial mass is equivalent to the gravitational mass for all objects. This would imply that the torque in part (b) disappears!

6. (10 points) **Thomas Precession** Suppose that students A and B start out in the lab frame. Student A remains in the lab frame. Student B gets in a rocket ship and changes their velocity by V_x in the x -direction (with $V_x \ll 1$) by firing the “+ X ” thrusters on their rocket. Then they change their velocity by V_y in the new y -direction (again with $V_y \ll 1$) by firing the “+ Y ” thrusters. Then they fire the “− X ” thrusters (changing their velocity by $-V_x$), and the “− Y ” thrusters (changing their velocity by $-V_y$). To lowest order in the V s, how does B ’s reference frame now differ from A ’s?

Note: The problem is designed so that in Newtonian physics, B ’s final reference frame is the same as A ’s: they are at rest with respect to each other, with no rotation of the coordinate systems.

Solution: We can write the Lorentz transformation matrix for going from A ’s frame (which is also B ’s initial frame) to B ’s final frame by multiplying the Lorentz transformation matrices. Remember that we apply matrices from

right to left:

$$\Lambda_{B_i \rightarrow B_f} = \Lambda(-V_y \mathbf{e}_y) \Lambda(-V_x \mathbf{e}_x) \Lambda(V_y \mathbf{e}_y) \Lambda(V_x \mathbf{e}_x) \quad (8)$$

$$\begin{aligned} &= \begin{pmatrix} \gamma_y & 0 & V_y \gamma_y & 0 \\ 0 & 1 & 0 & 0 \\ V_y \gamma_y & 0 & \gamma_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_x & V_x \gamma_x & 0 & 0 \\ V_x \gamma_x & \gamma_x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &\times \begin{pmatrix} \gamma_y & 0 & -V_y \gamma_y & 0 \\ 0 & 1 & 0 & 0 \\ -V_y \gamma_y & 0 & \gamma_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_x & -V_x \gamma_x & 0 & 0 \\ -V_x \gamma_x & \gamma_x & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \gamma_y \gamma_x & V_x \gamma_x \gamma_y & V_y \gamma_y & 0 \\ V_x \gamma_x & \gamma_x & 0 & 0 \\ V_y \gamma_y \gamma_x & V_x V_y \gamma_x \gamma_y & \gamma_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_y \gamma_x & -V_x \gamma_x \gamma_y & -V_y \gamma_y & 0 \\ -V_x \gamma_x & \gamma_x & 0 & 0 \\ -V_y \gamma_y \gamma_x & V_x V_y \gamma_x \gamma_y & \gamma_y & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -V_x V_y & 0 \\ 0 & V_x V_y & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} + \mathcal{O}(V^3) \quad (9) \end{aligned}$$

where in the last line, we have used the Taylor expansion of $\gamma_x \simeq 1 + V_x^2/2$ and $\gamma_y \simeq 1 + V_y^2/2$. The final Lorentz transformation corresponds to rotations around the z -axis by an angle of $-V_x V_y$ (recall that $\sin \theta \simeq \theta$ and $\cos \theta \simeq 1$ for small θ).

This rotation of the coordinate system when an observer boosts and comes back to their original velocity is called *Thomas precession*. It has no analogue in Newtonian physics.

One context where this is important is atomic physics: the spin of an electron orbiting an atom undergoes Thomas precession. A semiclassical description would be that as the electron orbits around many times, the direction of its spin angular momentum will precess around the direction of the orbital angular momentum. Since the typical velocity of an electron in a light atom is $\alpha = 1/137$ (the first Bohr orbit velocity in units where $c = 1$), we expect the precession to be by an angle of $1/137^2$ per orbit, or equivalently the precession frequency should be $1/(2\pi \times 137^2) \sim 8 \times 10^{-6}$ the orbital frequency – that is, about 50 GHz. This is a contribution to the fine structure splitting of

spectral lines or energy levels of atoms (11 GHz in hydrogen 2p; as one moves down the periodic table, the splittings are larger), but note that magnetic interactions give contributions that are the same order of magnitude.

The other context in which we care about Thomas precession is for space dynamics (satellites carrying gyroscopes orbiting the Earth, or binary stars). In the case of orbiting the Earth at a velocity of 7.5 km/s, or $2.5 \times 10^{-5}c$, we expect that each orbit yields a change in angle of

$$\pi \times (2.5 \times 10^{-5})^2 = 2 \times 10^{-9} \text{ radians} = 0.4 \text{ milliarcsec.} \quad (10)$$

A correct calculation for an orbiting satellite requires general relativity – the result is a factor of 3 different. In this context the phenomenon is known as *geodetic precession*. The measurement of geodetic precession by Gravity Probe B is one of the tests of general relativity.

7. (★) **The sky as viewed from a spaceship.** Let's suppose that observer $\mathcal{O}$ remains on Earth in the lab frame. A second observer $\bar{\mathcal{O}}$ moves in a spaceship at velocity $V = \tanh \alpha$ in the z -direction with respect to Earth. As you may recall from watching science-fiction movies, if V is large enough, $\bar{\mathcal{O}}$ sees the stars bunch up in front of them (the $+z$ direction). This problem works through the effect.

We suppose that the direction to the star makes an angle θ to the z -axis as seen from Earth, and $\bar{\theta}$ as seen from the spaceship. Without loss of generality, we will place the direction to the star at zero longitude (i.e., in the xz -plane).

- (a) (2 points) Show that in the Earth's frame, in time Δt , a photon from the star undergoes a displacement $\Delta x^\alpha = (\Delta t, -\Delta t \sin \theta, 0, -\Delta t \cos \theta)$.

Solution: The photon's time displacement is $\Delta x^0 = \Delta t$ by definition. Since it travels at the speed of light, the magnitude of its spatial displacement is Δt . The direction stated in the problem corresponds to the unit vector $(\sin \theta, 0, \cos \theta)$, but that is the direction the photon is from, so there is a $-$ sign. The result follows.

- (b) (5 points) Apply a Lorentz transformation to find the photon's displacement in $\bar{\mathcal{O}}$'s frame. You may leave some results in terms of $\gamma = 1/\sqrt{1 - V^2}$.

Show that in the barred frame, the direction of the photon satisfies

$$\cos \bar{\theta} = \frac{V + \cos \theta}{1 + V \cos \theta}. \quad (11)$$

Solution: The photon's displacement in the barred frame is

$$\Delta \bar{\mathbf{x}} = \begin{pmatrix} \gamma & 0 & 0 & -V\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -V\gamma & 1 & 0 & \gamma \end{pmatrix} \begin{pmatrix} \Delta t \\ -\Delta t \sin \theta \\ 0 \\ -\Delta t \cos \theta \end{pmatrix} = \begin{pmatrix} \gamma(1 + V \cos \theta)\Delta t \\ -\Delta t \sin \theta \\ 0 \\ -\gamma(V + \cos \theta)\Delta t \end{pmatrix}.$$

Now then $\cos \bar{\theta}$ is the $\bar{z}$ -displacement of the photon $\Delta x^{\bar{3}}$ divided by the spatial displacement in the barred frame (which is $\Delta x^{\bar{0}}$ because the photon travels at the speed of light in the barred frame), with an extra - sign because we want the direction to the star instead of the direction the photon is going:

$$\cos \bar{\theta} = -\frac{\Delta x^{\bar{3}}}{\Delta x^{\bar{0}}} = \frac{\gamma(V + \cos \theta)\Delta t}{\gamma(1 + V \cos \theta)\Delta t} = \frac{(V + \cos \theta)}{(1 + V \cos \theta)}. \quad (12)$$

- (c) (3 points) Show that a star that appears on the “Equator” as seen from Earth ($\theta = \pi/2$) has an apparent position $\bar{\theta} = \cos^{-1} V$ as seen from the spaceship. How far from the North Pole does the star appear in the spaceship frame if $V = 0.9c$? What about $0.99c$?

Solution: If we plug $\theta = \pi/2$ or $\cos \theta = 0$ into the previous answer, then we get $\cos \bar{\theta} = V$. This implies $\bar{\theta} = \cos^{-1} V$. Since $|V| < 1$, this is always a real number. The angle from the North Celestial Pole to the apparent position of the star in the spaceship frame is $\bar{\theta}$. By taking the inverse cosine, we see that this is 25.8° at $V = 0.9c$ and 8.1° at $V = 0.99c$.

- (d) (5 points) Now take the limit of small $\theta \ll 1$ (i.e., we will consider a constellation that contains the North Pole). Show that

$$\bar{\theta} \simeq \sqrt{\frac{1-V}{1+V}} \theta. \quad (13)$$

Hint: Take the Taylor expansion of your answer to (a) to 2nd order in θ . This means that the constellation containing the North Pole appears shrunk by a factor of $\sqrt{(1-V)/(1+V)}$ when seen from the spaceship.

Solution:

$$\begin{aligned}
 \cos \bar{\theta} &= \frac{V + \cos \theta}{1 + V \cos \theta} \\
 &= \frac{V + 1 - \theta^2/2}{1 + V - V\theta^2/2} + \mathcal{O}(\theta^4) \\
 &= \frac{V + 1 - V\theta^2/2 - 1/2(1-V)\theta^2}{1 + V - V\theta^2/2} + \mathcal{O}(\theta^4) \\
 &= 1 - \frac{1/2(1-V)\theta^2}{1 + V - V\theta^2/2} + \mathcal{O}(\theta^4) \\
 &= 1 - \frac{1/2(1-V)\theta^2}{1 + V} + \mathcal{O}(\theta^4). \tag{14}
 \end{aligned}$$

Note that in the last line, since the numerator is only nonzero to $\mathcal{O}(\theta^2)$, we can drop the higher-order terms in the denominator. We can then Taylor expand the left-hand side and find:

$$1 - \frac{1}{2}\bar{\theta}^2 = 1 - \frac{1/2(1-V)\theta^2}{1 + V} + \mathcal{O}(\theta^4). \tag{15}$$

With some minor manipulation, we end up with equation 13.